



Visual Search Tasks: The Effects of Head-Up Displays on Driving and Task Performance

Missie Smith
Virginia Tech
Blacksburg, USA
mis16@vt.edu

Jillian Streeter
Virginia Tech
Blacksburg, USA
jill229@vt.edu

Gary Burnett
Univ. of Nottingham
Nottingham, UK
Gary.Burnett@nottingham.ac.uk

Joseph L. Gabbard
Virginia Tech
Blacksburg, USA
jgabbard@vt.edu

ABSTRACT

This study investigated the differences between head-up display (HUD) and head-down display (HDD) systems in vehicles with the ultimate goal of increasing driver performance and safety. Our initial focus was on comparing drivers' visual task performance (and associated driving behavior) while using a HUD and HDD. In a medium-fidelity driving simulator, 16 experienced drivers performed two types of tasks, a structured (text) and a semi-structured (grid) visual search task while driving. They first completed a baseline drive using a driving simulator with no secondary tasks followed by four drives under various conditions: HUD-Text, HUD-Grid, HDD-Text, and HDD-Grid. The study returned mixed results. HUDs generally contributed to more secondary task errors, but overall, users still preferred to use HUDs over HDDs. The response time to complete tasks was unevenly affected by display type. While there was no difference between response times in the grid tasks (HDD vs. HUD), the HUD was associated with significantly faster task performance as compared to the text tasks. These results suggest that HUDs may offer advantages to drivers in some task scenarios but more research is needed to understand under what conditions they do not add value.

Author Keywords

Head-Up Display; Head-Down Display, Augmented Reality; Visual Display Image; Vehicle Display System.

ACM Classification Keywords

H.1.2 User/Machine Systems: Human Factors; H.5.m Information Interfaces and Presentation (e.g., HCI): Miscellaneous.

INTRODUCTION

Drivers often avert their eyes from the road to access vehicle-associated information such as speed, engine status, or entertainment controls to name a few. The risk of a traffic accident is significantly increased if a driver's gaze is away from the road for more than 2 seconds [1]. Car

manufacturers and display designers frequently seek new ways to provide useful information to drivers while lowering the mental demands associated with processing information. Positioning visual information closer to drivers' head-up, useful field of view (e.g., on or through the windshield of a vehicle) is one method of improving driving performance and lowering mental workload [2]. The expectation that driving performance can be improved using windshield displays has resulted in a higher interest in integrating information directly onto a windshield.

Augmented reality HUDs offer a promising mechanism to support this type of information integration possible. Augmented Reality (AR) blends the real world with virtual, computer-generated graphics resulting in the ability to place visual information and cues directly into users' line of sight [3]. Using graphics such as text, symbols or outlines allows users to experience the real world with digital enhancements that may improve their ability to perform tasks or gather information, such as identifying pedestrians or roadway hazards while driving.

Visual User Interface Design

In a vehicle, secondary information can be provided to drivers in many ways, two of which are in the form of head-up or head-down displays. A *head-up display* (HUD) is a display that provides information via digital graphics close to the driver's natural line of sight. A *head-down display* (HDD), in contrast, is a display which provides information below the driver's line of sight, usually on the dashboard or the middle console in a location which forces the driver to divert their visual attention away from the road [4]. AR HUDs have been shown to help keep drivers' eyes on the road with lower levels of mental workload, and consequently improved driving performance [5].

While AR HUDs are widely used in military aircraft allowing for easy access to flight information, navigation systems, and target locating [3], care must be taking in designing head-up user interfaces since visual complexity can dramatically affect the usability of HUDs [6]. Generally speaking, aircrafts have less visual background complexity due to environmental surroundings and fewer objects to interact with as compared to automobile settings. Incorporating HUD systems within ground vehicles is potentially subject to more risk due to highly dynamic traffic conditions and complex roadway navigation. In addition, there is debate as to whether adding graphics to the windshield will be beneficial or simply more distracting [7]. Finally, it is worth recognizing that the driver population is considerably more variable (in

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perceptual/cognitive abilities, expectations, motivations, etc.) compared to highly trained/regulated pilots.

While HUDs remove the need for a user to shift their gaze away from the road, AR images may change individuals' cognitive responses. Early research suggests that people cannot process the real world background and an AR graphic concurrently, as the mind has to treat them as two separate images [8]. Thus, simply overlaying information onto the real world can result in additional processing to, at a minimum, correctly identify what is real and what is computer generated, thus increasing the workload [9]. In driving scenarios, keeping eyes on the road may provide many benefits but the tradeoff of increased mental workload due to the need to parse out the real world from the digital may result in serious consequences.

One study of commercial drivers in Taiwan using HUDs and HDDs showed a decrease in response time to unexpected events and more consistent vehicle speed control, but no difference in average accuracy rate or mean vehicle speed when using a HUD [4]. Kerr's walking navigation study showed negative effects on performance when using AR displays [10]. For example, participants struggled to notice oncoming traffic when crossing roads or sidewalk trip hazards in Kerr's study. In addition, Kerr's participants frequently reported difficulty reading text due to bright light outdoors. At times, AR HUDs can improve performance, while at other times they clearly cause problems.

Task Design

The nature and structure of tasks must also be considered in analyzing HUDs. One type of secondary task that a driver might perform is a structured visual search task, such as reading a document or navigating a list arranged in alphabetical order. A user's search pattern follows a predictable path. Because of the nature of these tasks, it may be difficult to resume these tasks after interruptions. Individuals must be able to remember and located their exact spot after interruptions to be able to resume the task, so interruptions may result in re-starting the task or degraded performance.

An unstructured visual search task, in contrast, creates a visual field where the user's eye is not expected to follow one expected or set path during their visual search. The target might be positioned anywhere in the field, and does not have to stay on a line or grid pattern. For example, drivers should constantly scan the vehicle path for roadway hazards without any expectation of where one might occur.

Often, real world tasks are semi-structured, that is a hybrid of the structured and unstructured tasks. For example, a structured grid could be used to display visual cues, but the users would not have to search in the left-to-right, line-by-line method that is typical in structured visual search text tasks. In a vehicle, display lights can come on at any time (indicating low fuel, check engine, etc.), but there is a finite set of spaces where these lights can occur. Another example is the grid pattern of icons seen frequently in touch screen displays, such as smart phones. Semi-structured tasks are

typically easier to resume after interruptions, because locations can be generally remembered, rather than an exact location. Drivers perform a variety of search tasks, structured, unstructured, or a hybrid of the two, and it is important to understand how HUDs and HDDs affect different types of search tasks. While this study used an AR HUD specifically, the tasks were designed to be similar to those used in standard HUD displays.

Before AR HUD technology is used widely in ground vehicles, we must determine the differences between AR HUD displays and current technology. While keeping eyes on the road is an important factor, if AR HUD information is provided in a sub-optimal method, it may lead to confusion and further driver distraction. Using AR HUDs may provide benefits by keeping drivers' eyes on the road but there may also be hidden problems associated with adding more information to an increasingly complex system. This study examines the possible effects of using HUDs as compared to HDDs using two types of visual search tasks.

METHODS

Participants

Sixteen gender-balanced participants completed the experiment. The participants were 21 to 52 years old with previous driving experience in the United Kingdom and perfect or corrected vision (men: mean age = 28, SD = 5.2; women: mean age = 33, SD = 12.3).

Materials

The driving simulator was a fixed-base right-hand-drive vehicle. We used STISIM Version 2 software and a curved large screen projection to provide visual cues to drivers. A Pioneer HUD with a focal depth of about 3 meters was placed just above the driver's line of vision, roughly where a sun visor would be placed (Figure 1).

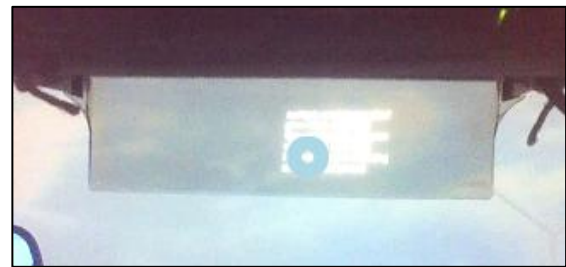


Figure 1: Pioneer HUD showing text visual search task. The thick blue circle over the text shows where the participant was looking as captured through eye tracking software.

A HDD was secured onto the middle console to the left of the driver, the standard location for vehicles in the United Kingdom. Only one display was used at a time to ensure that the participant knew where to look for task information.

The size and color of visual stimuli as presented via both displays were visually matched to appear as close as possible. Participants' eye gazes were recorded using an SMI eye tracking system with BeGaze analysis software,

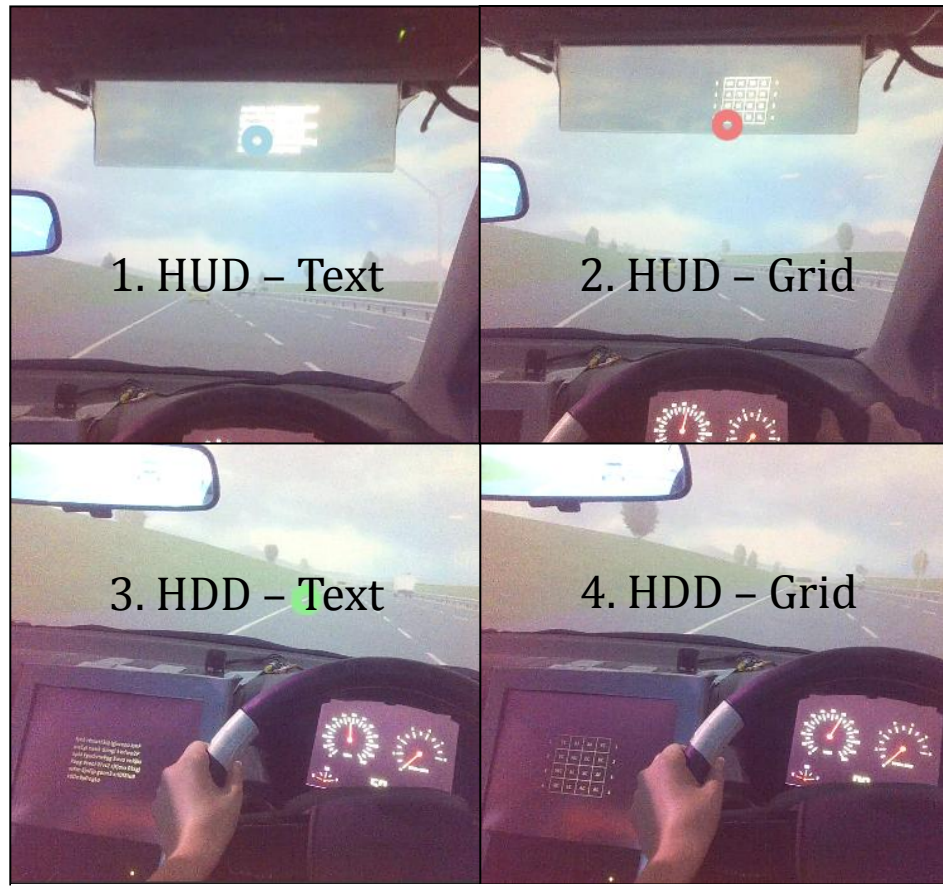


Figure 2: The four different display/image conditions were designed to simulate emerging displays and information formats.

although this analysis is not included in this paper. We used PowerPoint with custom Visual Basic for Applications software, embedded with time tracking, to control the images displayed on the HUD or HDD to participants, while simultaneously capturing response time data.



Figure 3: The HDD was fixed to the left of the driver, simulating a large tablet-style in-vehicle display.

Primary Task

The primary task of the participant was to drive a simulator vehicle on a standard UK motorway with 3 lanes in each direction, while following a lead car. The lead car fluctuated in speed from 65 to 75 MPH. Each participant completed the primary task six different times, including one familiarization drive, one baseline drive, and four different driving scenarios in which they also performed a secondary visual search task (described below).

Both degraded performance and more conservative driving habits are indicative of an increase in mental workload [11].

Two types of metrics are commonly used in the evaluation of driving performance: longitudinal control and lateral control. Longitudinal control is evaluated according to SAE J2944 standards by measuring time to collision, distance or time gap, distance or time headway, and other similar measures. Lateral lane position, measured in feet, is also used to assess driving performance [12]. For the purposes of this study, lateral lane position was defined according to SAE J2944 10.1.1.3 (Option C). The participants drove on a flat and straight road, diminishing common concerns with using Option C in lane position evaluation of driving simulators. The Standard Deviation of Lane Position (SDLP) as defined in SAE J2944 10.1.3 was used for all SDLP calculations (Equation 1).

$$SDLP = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$$

Equation 1: SDLP Calculation, where $x_i = i^{th}$ value of lane position, $\bar{x}$ = mean lane position of the sample, N = Number of data points in the sample

Secondary Visual Search Tasks

Throughout the study, participants completed two different types of secondary tasks: structured and semi-structured visual search tasks. We visually matched the color and size of text between the two displays to ensure that the images were as close as possible. We changed the color of the

HDD graphics to mimic the slight green tint of the HUD. Because the matching was done visually rather than strictly controlling for size, the font and colors may have been slightly different.

Structured Visual Search Task (Text):

For this study, pseudo text was used in place of familiar words in order to avoid influence from a user's reading pace or previous knowledge of words or context. The pseudo text is built from a randomized sequence of letters, both capitalized and lowercase, spaced similarly to written words but following ISO 9241-3 [13]. To complete the visual search task, a participant scans this pseudo text to locate and count the occurrences of a pre-determined target letter 'E' (Figure 3). Each display had at least one target letter and no more than six. There were three levels of task complexity with 2, 4, or 6 rows of text for low, medium, and high difficulty tasks, respectively. After reading through the text, participants verbally stated the number of target letters present in that specific set of pseudo text.

g3X7Pj eBsgil i8jzGsAT vPdk
Tyl e9Xfm 04lj RbEGGP khub

Figure 4: This pseudo-text visual search task forces bottom-up processing, requiring participants to focus on each letter to locate and count targets (capital 'E'). The correct answer for this set is 1.

Semi-structured Search Task (Grid):

To conceptually simulate a 2D indicator/control layout (e.g., that could be used in center console display), we developed a square grid layout where participants would need to locate a set of target letters (i.e., 'AC') within a $n \times n$ pattern of visual stimuli.

1	WC	BC	XC	DC	1
2	GC	LC	JC	AL	2
3	SC	AC	RC	HC	3
4	KC	EC	FC	OC	4

Figure 5: HDD Semi-Structured Visual Search Task, 4x4 Grid. Due to the design of the HUD, the text had a green tint to it.

While a left-to-right reading pattern could be followed, the grid structure of the visual task allowed for a variety of eye movements. The location of the target AC within the grid was randomized. Each grid cell contained two-letter combinations selected from a pool of potential distractors either beginning with 'A' or ending in 'C'. No real words

(e.g. AT, AM) were used in the grid. To complete the task, participants had to first locate, then verbally state the row location of the single target. We used 3x3, 4x4, and 5x5 grids for low, medium, and high task complexity (respectively).

Procedure

Prior to any driving, participants familiarized themselves with the displays and secondary tasks. They then completed the practice and baseline drives. Participants experienced the four secondary task conditions one at a time in a counterbalanced order. At the beginning of each drive, participants began by engaging in the primary task of following the specified car. Once underway, we introduced the secondary task demands. Each participant completed 15 trials for each condition (3 levels of complexity x 5 replications) with a random interval between 10-15 seconds of driving time between each trial. To indicate completion of the trial (counting the 'E' targets for the text condition or stating the row number of the 'AC' target in the grid condition), participants verbally stated the answer, and subsequently reported their confidence in performing the secondary task (ranging from 1, lowest confidence, to 10, highest confidence). Participants stopped the car after completing the last trial.

We recorded each answer and confidence level, and later manually calculated the number of errors for each condition. As soon as a participant stated the number or location of targets, we advanced to the next slide. The custom Visual Basic for Applications software logged these times, which we then used to determine the amount of time required to complete each task. We tracked participant eye movements such as glance number, glance length, and fixation time using the SMI eye tracking suite. For the primary task, speed and lane position were captured by the STISIM driving software. After completing all drives, we asked participants which display they preferred for legibility, ease of use, convenience, and overall. They were then asked to complete NASA-TLX rate 6 components of workload under each condition.

ANALYSIS

For this paper, we report on dependent measures related to primary task (driving) performance (lane position), secondary task performance (task response time and error) and participant preferences related to the displays.

Driving Performance Data

While we have collected a suite of longitudinal data, we have not yet examined SAE-recommended longitudinal data such as time to collision, distance to vehicle ahead, distance gap, time gap, etc., but did an initial look at speed and standard deviation of speed. We excluded the first 2,000 feet of driving distance because this was the space required to merge onto the road, steady the car, and begin to follow the lead car. All data after participants slowed down from 4th gear for the last time was also excluded, as this indicated that they had finished the secondary task and were instructed to stop the vehicle to end that condition.

Because the driving simulator data collected was not normally distributed, we used Friedman's test to examine differences between the mean and standard deviation of (1) vehicle speed and (2) lane position in the control and four experimental conditions. Only SDLP showed a significant difference ($Q=13.55 > \chi^2(4, 0.05)=9.488$). We further analyzed the standard deviation of lane position to determine where these differences laid.

Condition	Mean SDLP	St. Deviation of SDLP	Median SDLP
Control	0.93	0.24	0.92
HUD-Text	1.55	2.05	1.05
HDD-Text	1.14	0.28	1.05
HUD-Grid	0.92	0.24	0.84
HDD-Grid	1.00	0.22	0.99

Table 1: Mean and Standard Deviation of SDLP

Specifically, we performed Wilcoxon Signed Rank tests on each of the standard deviation-lane position pairs and found the following significant differences (Table 2). The other pairwise comparisons not listed were not significant.

Pair	Z-score	P-Value	Higher SDLP
Control vs. HUD-Text	2.017	0.044	HUD-Text
Control vs. HDD-Text	2.379	0.017	HDD-Text
HUD-Text vs. HUD-Grid	2.327	0.020	HUD-Text
HDD-Text vs. HUD-Grid	2.637	0.008	HDD-Text

Table 2: Independent of display type, the text conditions were commonly associated with higher standard deviations of lane position.

There was no difference in the SDLP for HUD and HDD for either the text ($Z\text{-score} = 1.034$, $p\text{-value} = 0.301$) or grid tasks ($Z\text{-score} = 1.138$, $p\text{-value} = 0.255$). Therefore, the HUD did not have a significant effect on driving performance.

Response Time

As expected, secondary task response times were not normally distributed, nor was the commonly used log transform of response times (although it nearly met the criteria). As such, we used the Wilcoxon Signed Rank test and observed no significant differences between response times across the two display types in the grid visual search task condition ($p=0.081$, $Z=1.396$). We did however see a significant difference between the HUD and HDD in the text visual search condition ($p=0.002$, $Z=2.947$). This suggests an interaction effect between the display type (HUD or HDD) and image (Grid or Text) presented to the user (Figure 8).

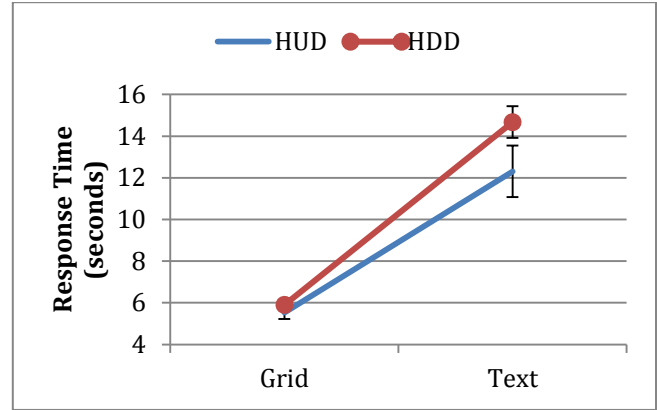


Figure 6: There is likely an interaction effect between display type and image type.

Errors

Incorrect row numbers (grid conditions) or target counts (text conditions) were counted as errors. Errors by participant were analyzed using nonparametric methods because while the data was normally distributed, there were only 16 responses for each category. HUD usage was associated with more errors in both the text and grid visual search task conditions, with the vast majority of errors occurring in the text conditions. A Wilcoxon Signed Rank test indicated that using the HUD resulted in a slightly higher error count as compared to the HDD ($S = 20.5$, $p = 0.0847$).

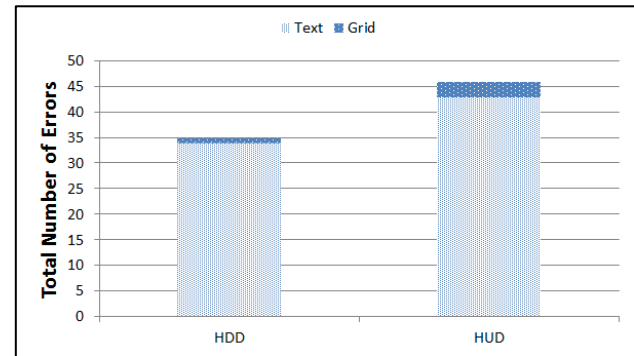


Figure 7: Few errors were associated with the Grid tasks.

Display Preference

Overall, participants preferred head up displays for both the text (78%) and grid visual search tasks (80%). However, the legibility of the HDD-text display was preferred over the HUD-text display (63%). This is congruent with Kerr's findings in a study considering AR personal navigation devices where participants reported that they were unable to effectively read text in HUDs, but that they generally liked using the AR devices [10]. Another possible explanation for the lack of legibility in the HUD is the inherent interaction between HUD text and driving scene backgrounds as observed by Gabbard, Swan & Hix [14] as well as Gattullo et al. [15].

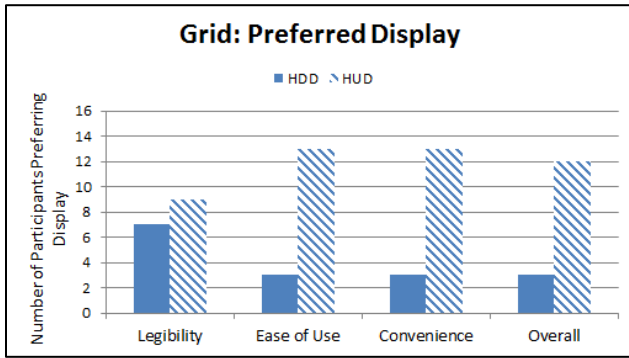


Figure 8: The Grid tasks were associated with a strong preference for using HUDs.

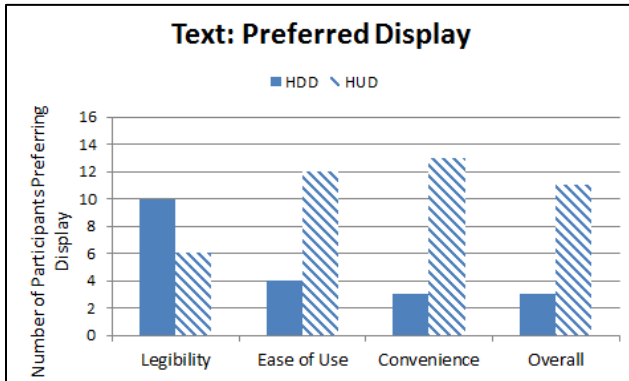


Figure 9: The legibility of the Text task was compromised when used with a HUD.

With the exception of legibility, the HUD was strongly preferred by participants in both grid and text tasks.

Confidence Level

Confidence levels were not normally distributed so we used the Wilcoxon Signed Rank test to examine individual differences. Confidence levels were significantly different across the text and grid visual search tasks ($Z=3.408$, $p=0.0003$). A similar analysis indicated that the HUD and HDD displays were associated with different, but not statistically significant, confidence levels ($Z=1.726$, $p=0.0843$).

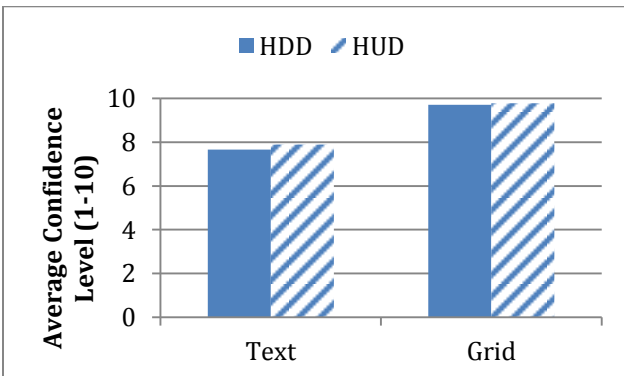


Figure 10: Both display (HUD & HDD) and image type (Grid & Text) affected confidence levels.

NASA Task Load Index (NASA-TLX)

We performed Friedman's test to determine if there was a difference between the reported mental workload between the four display conditions. There was a significant difference ($Q=18.075 > \chi^2(3, 0.05) = 7.815$), so we performed Wilcoxon Signed Rank pairs for each of the display condition pairs. We found the following significant differences (Table 3). The other pairwise comparisons not listed were not significant.

Pair	Z-score	P-Value
HUD-Text vs. HDD-Grid	2.706	0.007
HUD-Text vs. HUD-Grid	2.950	0.003
HDD-Grid vs. HDD-Text	3.180	0.002
HUD-Grid vs. HDD-Text	3.181	0.002

Table 3: Pair-wise comparisons of subjective mental load.

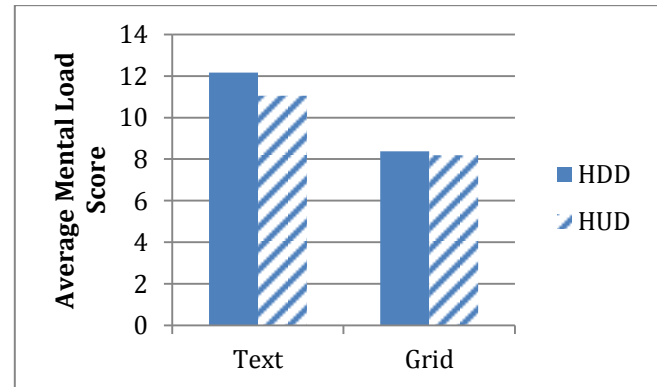


Figure 11: Average reported mental load was higher for text.

DISCUSSION

This study paints an initial picture of driver performance and preference considerations when designing for in-vehicle HDD and HUD displays. While HUDs offer some potential benefits such as perceived increased ease of use and convenience for drivers, they may result in compromised driving performance such as an increase in errors for specific secondary tasks.

Task Design

We expected that the text visual search task performance would suffer as participants would find it difficult to resume this task after interruption (i.e. looking back at the road to ensure safe driving). The higher number of errors, lower confidence levels, and higher reported mental load all support this expectation. Some of the differences in errors, confidence, and mental load may also be related to inherent task difficulty. Conversely, the grid visual search task, which was designed to better support resumability, scored higher in each of these areas.

In addition, the text visual search tasks were associated with higher standard deviations of lane position as compared to the grid visual search tasks and the control. This indicates that there was more visual distraction in the text tasks [16].

Task design also affected the type of lessons learned that can be gleaned from our study. The difference in error rates between this study and Liu's [4] may be due to the nature of the tasks performed. Liu's experiment used commercial goods delivery professionals and incorporated secondary tasks that were directly related to the driving and delivery tasks. Our study sought to understand how adding cognitive load via HDD and HUD would affect driving and secondary task performance on ordinary drivers when participants could not use logic and context to infer likely answers. Instead, our secondary tasks purposefully examined bottom-up cognitive processing to provide a glimpse at the basic level on the effect of adding various cognitive loads might have on drivers.

Visual User Interface Design

Even with the potential benefits of HUDs, moving forward we should be careful in designing information displays for drivers because some HUD interface designs may be associated with greater degrees of lane drift than others. Therefore, the design of the information flow (i.e. top-down vs. bottom-up) should be carefully considered in conjunction with display type. For example, Tables 1 and 2 suggest that a text visual search task (bottom-up) may be associated with greater lane drift than grid-based visual search tasks (top-down). What are the effects of other types of visual information processing tasks on lane drift?

People may feel more comfortable using a HUD, as evidenced by the higher confidence levels and preference scores observed. This is congruent with Kerr's findings that participants were unable to effectively read text, but that they generally liked using the AR devices [10]. Our results suggest that comfort and tendency to accept new forms of technology are not directly associated with improved performance. This conflict in preference and performance is often cited in usability studies (e.g., [17]). In this instance, both driving performance and secondary task accuracy were worse in HUD-text conditions, even though participants generally preferred the HUD and reported no increase in mental workload in these conditions. Importantly, our study provides evidence that individual preferences such as willingness to readily adopt new technology might predispose drivers to use HUDs even though it may diminish their driving abilities.

Furthermore, the legibility problems noted by participants pose some concern in the HUD condition. The text condition had less space between text, which we think is closely connected to the difference between legibility preferences of the HUD and HDD conditions. The degradations in performance (errors, speed) in the HUD-text condition may be due to the text being hard to read. Using a more legible text could have resulted in speed improvement with similar or fewer errors to the HDD-text condition.

In addition, see-through AR HUDs are typically subject to color blending [18] which can affect the overall usability of

the display. Increasing the brightness of a display can create richer looking colors and minimize blending, but may in turn wash out the display images [19] and/or occlude important real-world background visual cues. We believe the HDD display was considered more legible because of this washing out of the display image in the HUD condition. However, even with a clearly superior HDD, participants preferred the HUD overall. Participants were biased in favor of new technology regardless of its effect on their performance.

CONCLUSION & FUTURE WORK

While some task designs (i.e. structured, non-resumable) are not well-supported by HUDs, people seem predisposed to adopting this technology without realizing how much it distracts them from the driving task. Performance may be affected even though the images are displayed within the line of sight. Other displays of information, such as grid patterns that do not afford an obvious visual search pattern, but support resumability, may be better suited for HUD presentation (assuming that the interface maintains a reasonable level of visual complexity).

Display designers need to carefully consider the high acceptance of HUDs due to novelty before implementing interface designs, and they should develop interfaces that that have been proven to support resumability. Because people perceive that they are still paying attention to the primary task of driving, HUD tasks that do not support resumability (and therefore encourage drivers to focus on the display for longer periods of time) may cause even bigger problems than HDD interfaces. In this case, an interface that cannot support resumability should not be incorporated into HUDs at all.

Future work will re-evaluate this data considering the longitudinal measures of driving data according the SAE standards. In addition, we will examine the data collected about eye-tracking to understand more about the visual attention of drivers when using HUDs and HDDs in various types of visual search tasks. Task complexity with relation to display type should also be examined. Further studies must be performed in order to understand the intricacies of the interactions between display type and the information or images presented to drivers.

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